

Figure 1: Architecture for Big Data analysis (Image credits: googleimages.com)

## How large companies use Big Data

1. The techniques listed later in this article help to extract useful insights from large data sets, which are leveraged further for different surveys, statistics and case studies.
2. Flipkart, Amazon and other such online e-commerce sites make use of these techniques to study the behaviour of their customers and help them get what they want.
3. Facebook supposedly knows more about each one of us than our own therapists do and it's possible only because of the different Big Data techniques it implements. It continuously keeps track of different user actions, photo uploads, comments, shares, etc, using these techniques.
4. MNCs like Walmart make use of Big Data to improve their 'employee intelligence quotient' and 'customer emotional intelligence quotient'.
5. Family restaurants like Dominos, McDonald's and KFC use predictive and user-behaviour analytics to increase the efficiency of their marketing and continuously improve the customer experience.

## What must be considered while analysing Big Data?

Let's have a look at the different factors that an analyst must consider while working with Big Data.

1. **Ingestion:** This process is about moving data (especially unstructured data) from where it originated, into a system where it can be stored and analysed. Data ingestion can be continuous or asynchronous, in real-time or batched, or even both.
2. **Harmonisation:** This deals with the improvement of data quality and its use with the help of different machine learning capabilities. It also interprets the characteristics of the data and the different actions taken on it, subsequently using that analysis to improve data quality.
3. **Analysis:** This deals with the analysis of the data sets in order to understand the behaviour of data and identify specific patterns or trends, so that different actions can be



Figure 2: Use of Big Data analysis in the banking sector (Image credits: googleimages.com)

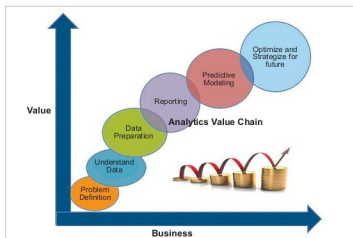


Figure 3: Increase in business value due to Big Data analysis (Image credits: googleimages.com)

- performed on that data set.
4. **Visualisation:** Data visualisation is the process of presenting the data in a pictorial or graphical format. This helps decision makers to grasp difficult concepts or identify new patterns in the data set.
  5. **Democratisation:** This is the ability for specific information in a digital format to be accessible to the end user. This is used to enable non-specialists to gather and analyse larger data sets without requiring outside help.

## Popular techniques for analysing Big Data

Let's have a look at some of the popular techniques that are being used for analysing large data sets. All of them generate useful insights that can be used further for diverse applications.

**A/B testing:** This is a method of comparing the two versions of an application to determine which one performs better. It is also called split testing or bucket testing. It refers to a specific type of randomised experiment, under which a group of users is presented with two variations of some product (an email, advertisement or Web page)—Variation A and Variation B. The users exposed to Variation A are referred to as the control group, because